BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number: BIT/2024/54480

School/Faculty: Computing and informatics

Programme: _Mobile application development

Academic Year: 2026

Semester: ii

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Week 1: Introduction to Mobile Application Development & Week 2: Mobile Development Environment

Practical work done

Android studio environment set up

Setting up first android studio project

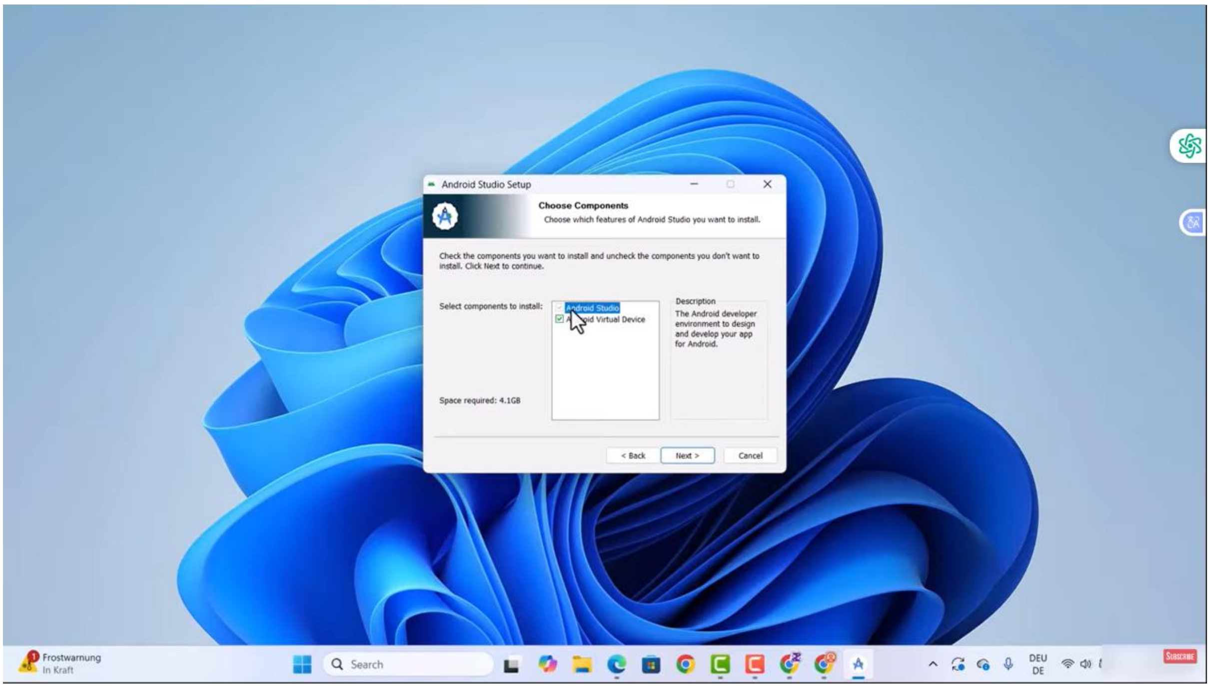
Learned how frameworks work and how UI/UX can be used

Learned how AI Agent works in referencing the code in the code window

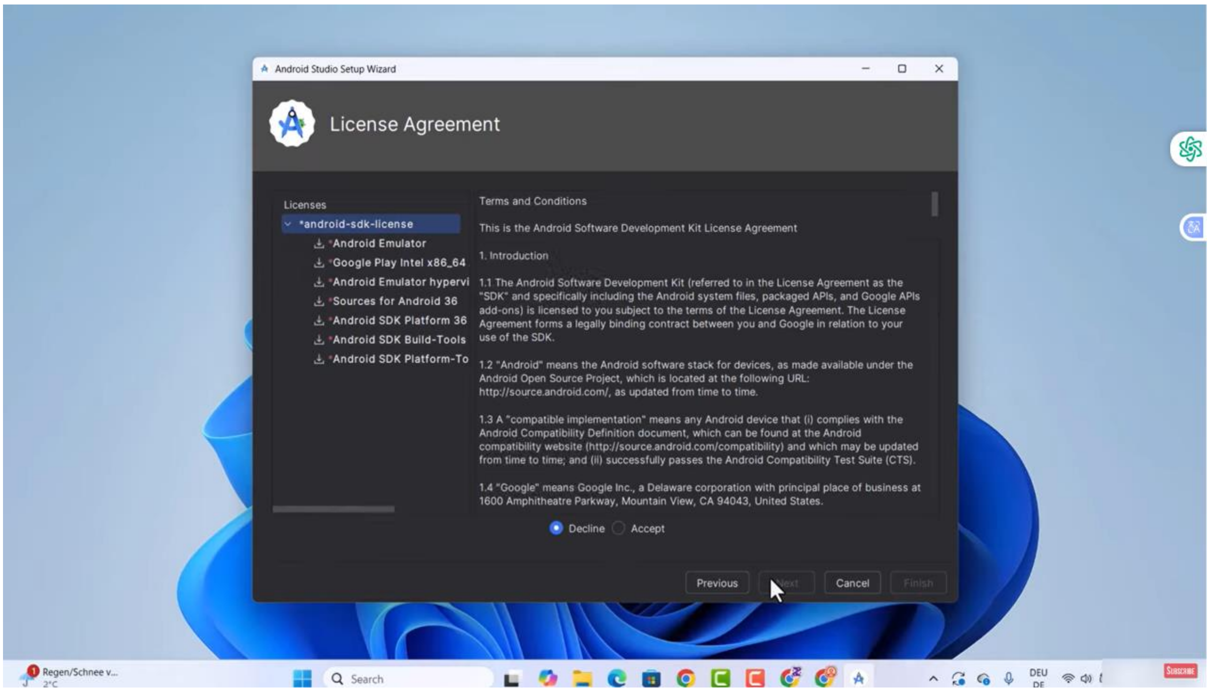
Learned how to use some basic flutter language

The screenshots below will show evidence of the processes that i followed up to the latest updates in GitHub

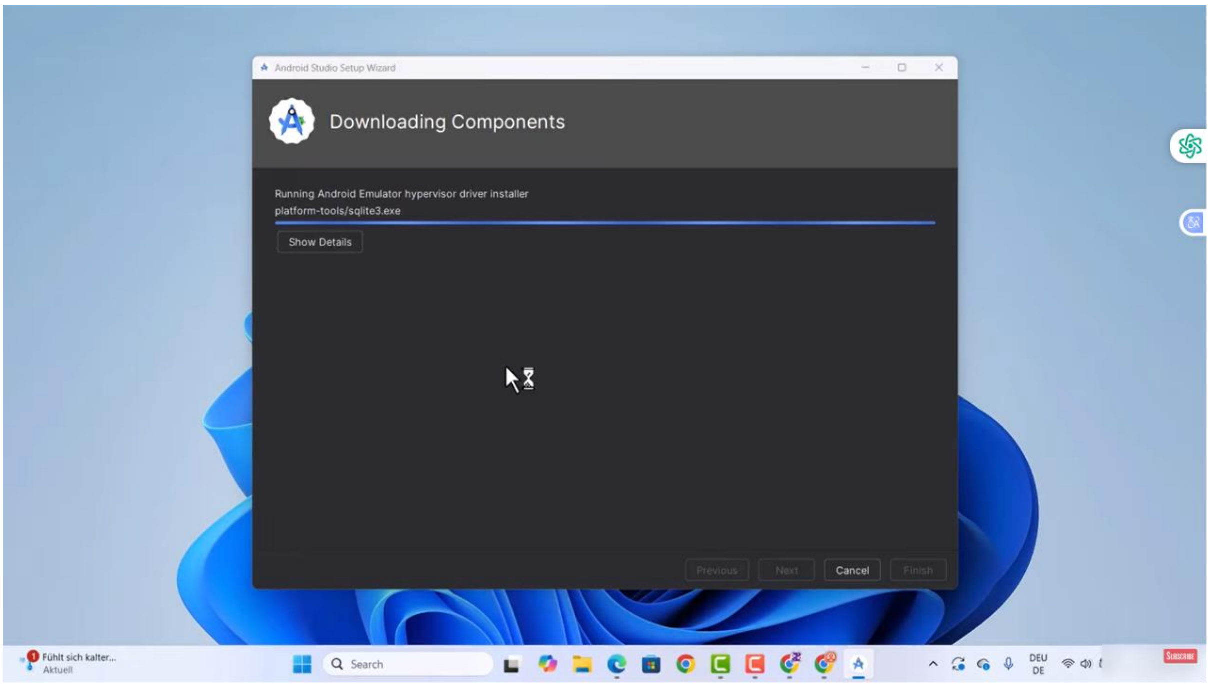
Android studio installation



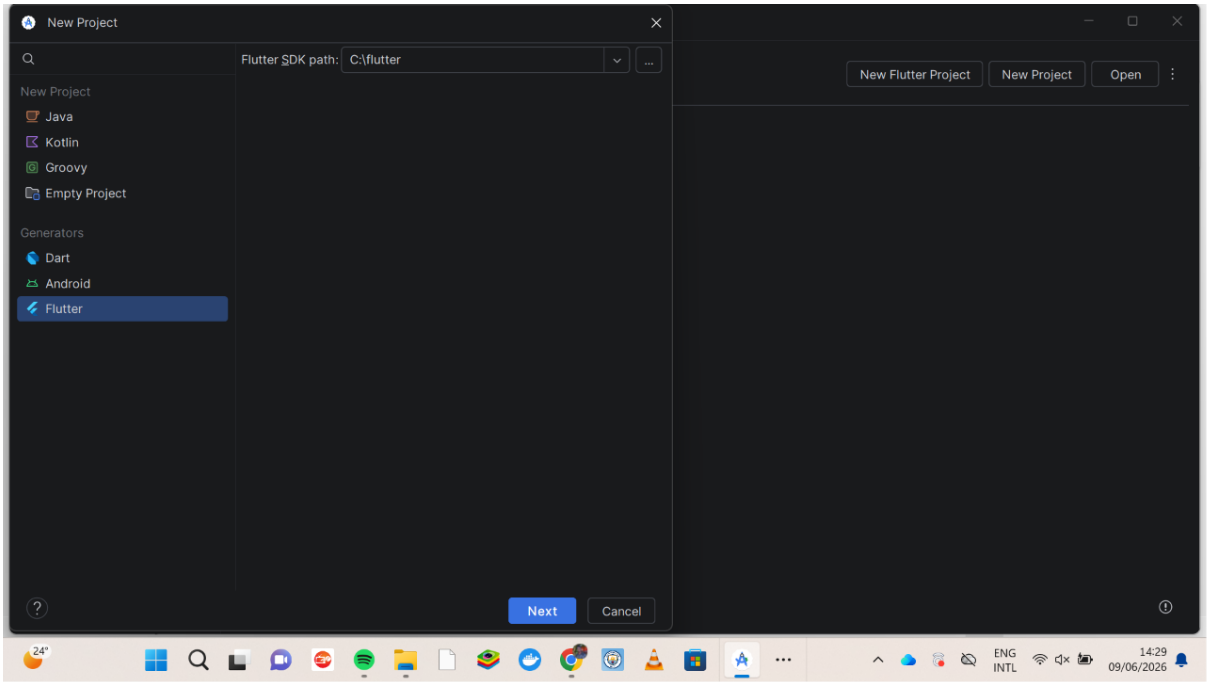
Licence agreement setups

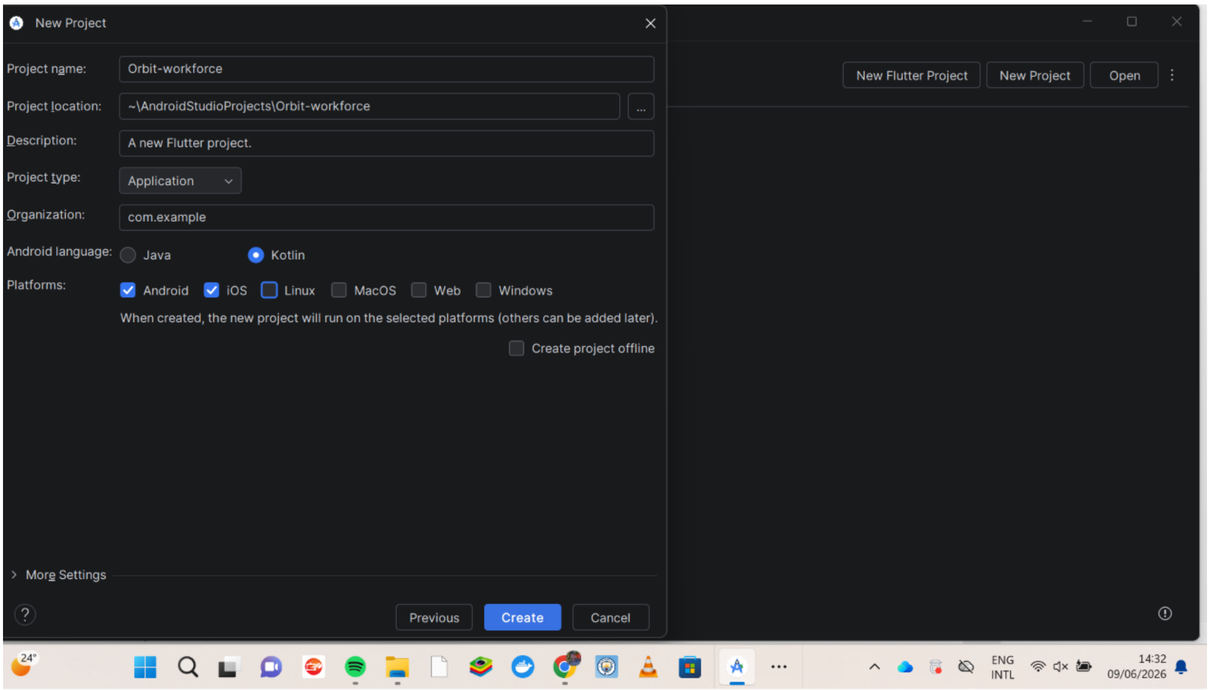


Components dowloads

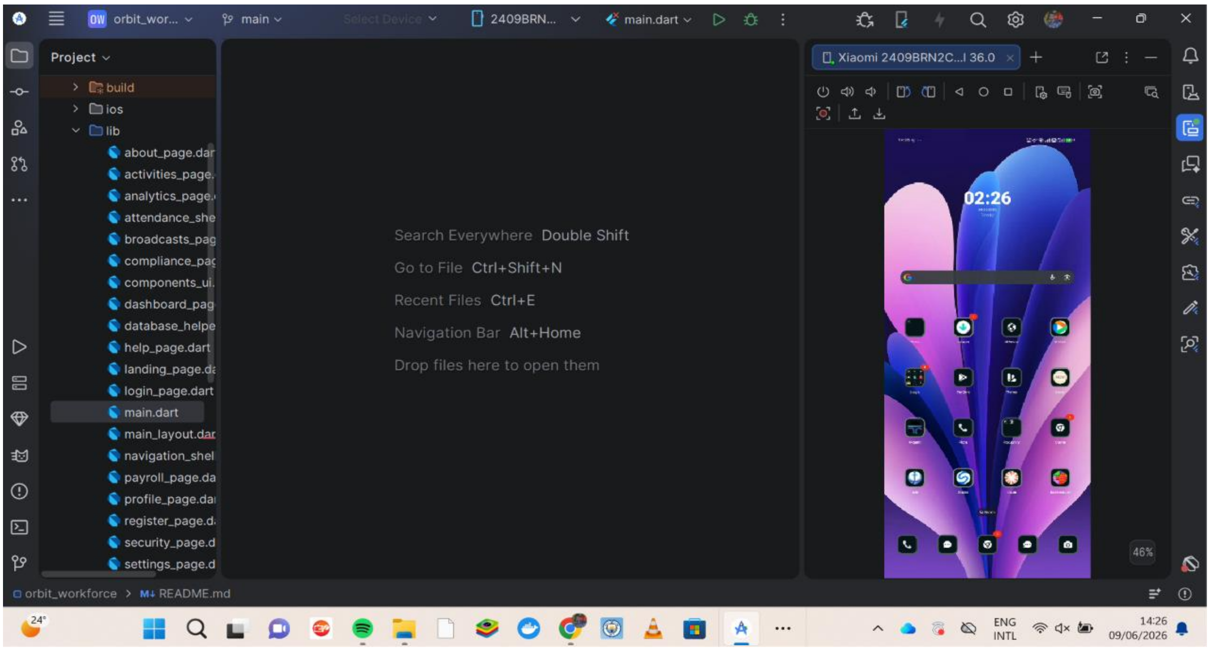


starting new project





Setting up an external phone to be used as an emulator



Tools used for week 1 and week 2

Android studio

Flutter SDK

Flutter plugin

Dart plugin

My personal reflection was how i was able to understand how applications development success really depend on how you start the essential steps at the beginning.

To my own experience setting up the environment wrongly result in much disappointment that may result in stagnation and halt progress definetly.

Week 3: User Interface Design

Task: Development of the Orbit Workforce Landing and Attendance Interface

Design Approach I prioritize a mobile-first approach because I design for workers on the move. I use a clean, professional aesthetic to build trust and ensure the application remains readable in outdoor environments with high glare.

Key Design Decisions

- I implement a consistent visual hierarchy by using large, bold typography for primary actions like "Check-in." This makes it easier for users to interact with the app quickly.
- I design the site selection area using Card widgets. I do this to create clear boundaries between different job locations, preventing user errors when selecting a site.
- I maintain brand consistency by choosing a primary blue color palette, which I find conveys a sense of professionalism and reliability.

Technical Implementation

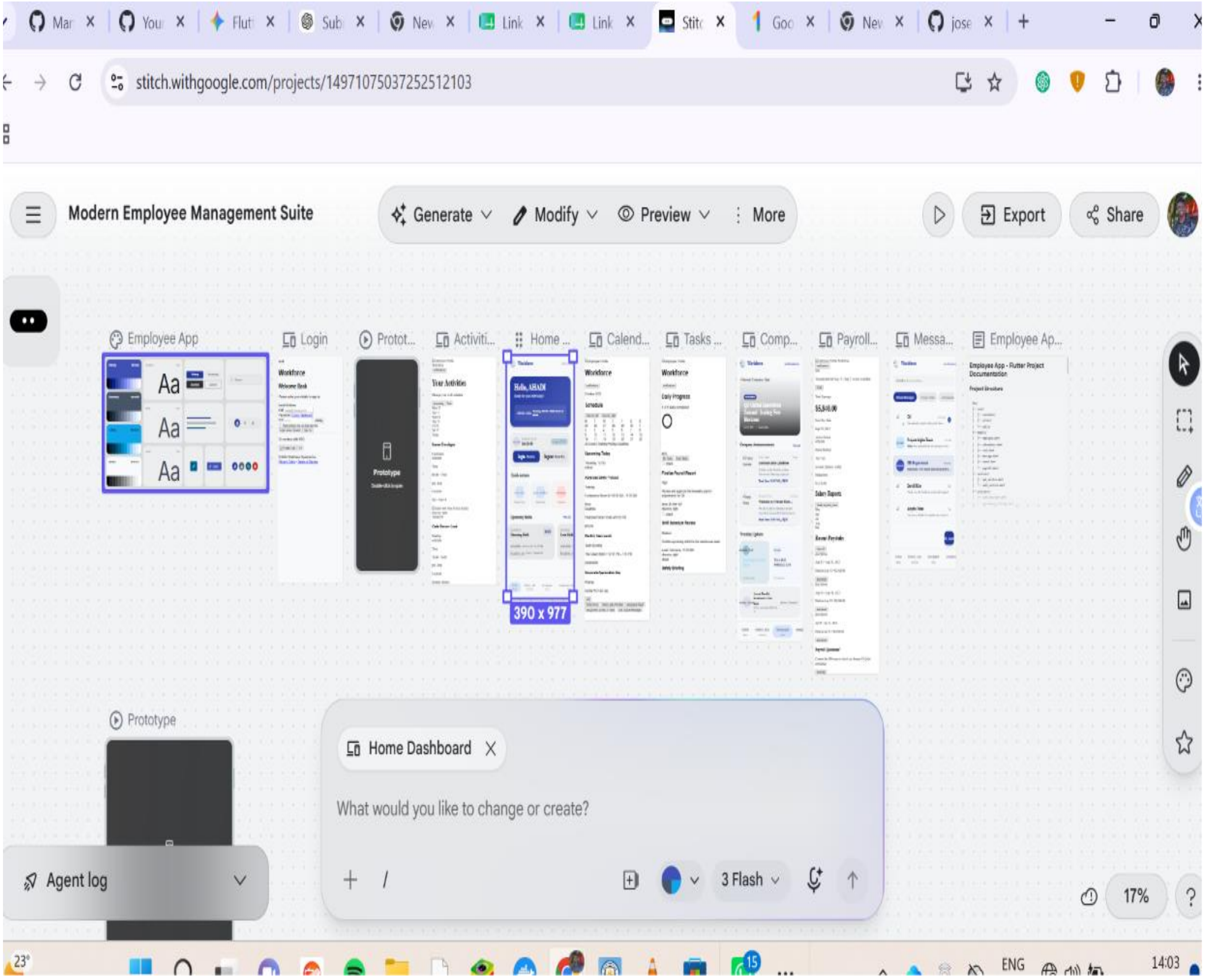
- I use the Scaffold widget to provide a stable, standard structure across every page in the app. This ensures that the navigation remains in the same spot, which helps users learn the interface faster.
- I bind the UI to the database by using ListView.builder. I do this to ensure that attendance logs update dynamically whenever a user performs an action, providing them with instant feedback.
- I define all my colors and font sizes in a central theme file. I do this so that I can update the entire app's look and feel from a single location, which saves me time during the development process.

Refinement Process

- I conduct quick usability checks by navigating through the app myself. If I notice a button is hard to press, I increase its hit area to 48x48dp.
- I simplify the registration flow. I notice that too many input fields cause friction, so I remove non-essential requirements to ensure the user reaches the dashboard as quickly as possible.

I USED STITCH TO IMPLEMENT ALL THE USER INTERFACE

DESIGNS



After using the Stitch frame work it comes with inbuilt code that can be used in different languages, for my case i had to convert that language to dart so that it can work with flatter effectlively.

- **Week 4: Event Handling and User Interaction**

  I focus on creating an intuitive experience where every user action triggers an immediate and predictable response from the application.

  Steps I Follow for Implementation

1. **I identify the trigger:** First, I determine which user action initiates the event (e.g., a button press, a form submission, or a list item tap).
2. **I define the callback function:** I create a function that houses the logic I want to execute. For example, when a user clicks "Check-in," **I write** an asynchronous function that calls my DatabaseHelper to save the attendance log.
3. **I implement the UI listener:** I wrap my interactive widgets (like ElevatedButton or InkWell) in the appropriate event listener parameters, such as onPressed or onTap.
4. **I manage the state:** Because the app needs to update after an interaction, **I use** setState() or a state management solution. **I trigger** a UI rebuild so the user sees the change immediately (e.g., showing a "Success" message).
5. **I handle user feedback:** I ensure the app tells the user what happened. **I display** a SnackBar or a dialog box to confirm the action, which keeps the user informed and prevents uncertainty.
6. **I include error handling: I wrap** my interaction logic in try-catch blocks. If the database fails or the internet is slow, **I provide** a clear error message to the user instead of letting the app crash.

   <u>Below is a flowchart that vividly describes how each steps i took in ensuring that all the events to be handled will be effectively synced</u>

Week 5: Data Management

Week 6: Networking and APIs

Same as week 1

Week 7: Application Architecture

Same as week 1

Week 8: Project Planning

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic

Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: ___________________________

Student Signature: _______________________

Date: ___________________________________

Lecturer Name: __________________________

Lecturer Remarks: ______________________

Date: ___________________________________